

Introduction

MOTIVATION

- A visual body cue is not a verified emergency. Viewpoint, scale and appearance shift substantially from benchmark data to a bedside robot camera.
- One accuracy value hides the two outcomes that decide the interaction: a false positive escalates for nothing; a silent miss stops the robot engaging at all.
- Confirmation-before-alert is established in fall detection; what is missing is evidence across a multi-cue distress vocabulary.

MAIN CONTRIBUTIONS

- We quantify benchmark-to-robot transfer over a multi-cue distress vocabulary, comparing RGB and pose-centric paths on the same 28 robot-camera clips.
- We separate exact cue recognition from interaction-relevant outcomes: question triggers, silent misses, unnecessary questions and waving responses.
- We validate the question-first controller with 13 event-injection trials, isolating control reliability from recognition accuracy.

System Design

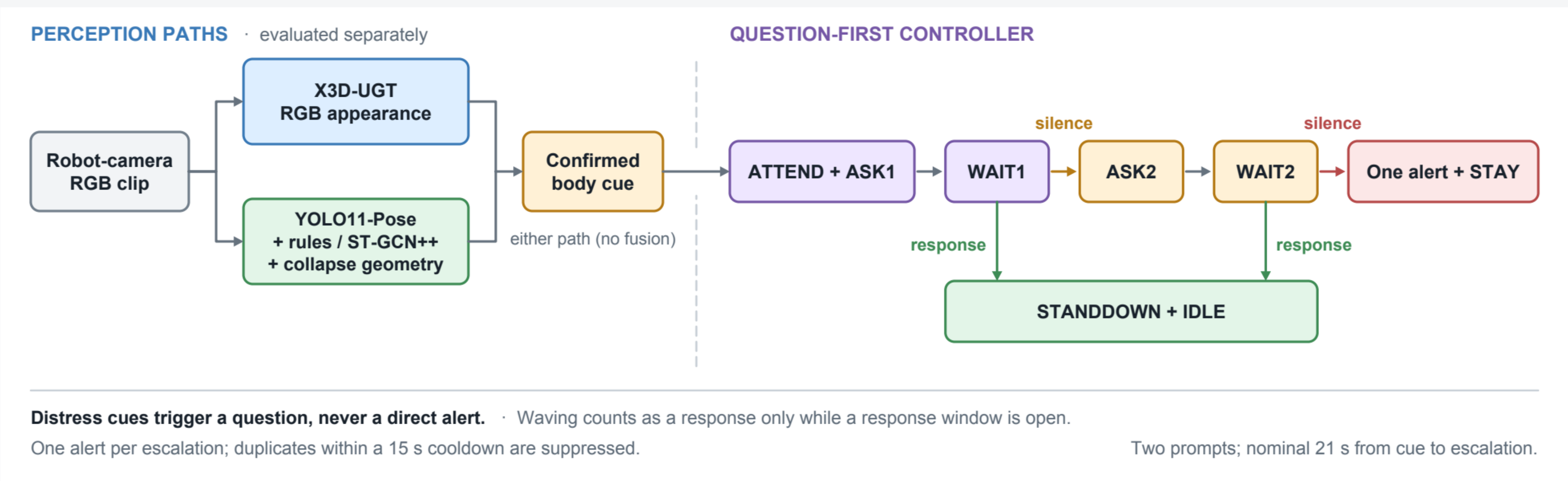


Fig1. Perception paths and the question-first controller. A confirmed distress cue opens a response window instead of an alert.

QUESTION-FIRST INTERACTION

- A confirmed cue makes Nuni attend and show a text prompt for 2.5 s, followed by an 8 s response window.
- A response stands the system down. Continued silence issues a second prompt and window, and only then publishes one alert — nominal 21 s, with a 15 s cooldown.
- Nuni is a bedside prototype on a commodity pan-tilt camera, with a Jetson Orin Nano as the on-device deployment target.

TWO PERCEPTION PATHS

- RGB appearance: X3D-UGT, a sub-1M-parameter RGB-only network at 95.1% / 90.9% top-1 on NTU RGB+D 60/120, in two six-way variants — fine-tuned (97.7%) and from scratch (94.8%).
- Pose-centric hybrid: YOLO11-Pose keypoints, gesture rules, a 48-frame ST-GCN++ component and bounding-box transition geometry for collapse.
- Persistence and debounce turn window predictions into confirmed tags; the two paths are evaluated separately and never fused.

Experiments

PROTOCOLS

- 28 trimmed, single-actor scripted clips from the robot camera: four per distress cue (16), four waving and eight normal, each held ~8 s at 1–2 m. RGB paths were scored more leniently than the pose path, so the gap is conservative.
- For the controller, the camera and recognition are bypassed and ROS 2 events are injected, isolating control reliability from recognition.

Perception Results

BENCHMARK-TO-ROBOT TRANSFER

Perception path	NTU acc.	Macro	Exact	Question	Silent
Pose-centric hybrid	89.8%	0.71	11/16	12/16	4/16
X3D-UGT, fine-tuned	97.7%	0.25	1/16	2/16	14/16
X3D-UGT, from scratch	94.8%	0.29	3/16	3/16	13/16

Table1. Clip-level outcomes on the 28 robot-camera clips.

- Benchmark accuracy did not predict robot behaviour: between the two directly comparable RGB variants the ordering inverts. The hybrid kept a question-triggering cue in 12/16 distress clips, the RGB variants in 2/16 and 3/16.
- Per class, gross motion transferred best — staggering 4/4 for the hybrid — while both RGB variants scored 0/4 on chest, headache and staggering. A crop ablation rules out framing: RGB macro recall stayed at 0.17–0.33.

Controller Results

- All 13 event-injection trials passed: five responses stood the system down, five silences raised exactly one alert each, and three boundary cases behaved as specified.
- No duplicate alerts, deadlocks or crashes. Cue to the first response window 2.62 s; stand-down within 0.19–0.20 s; one alert after 21.65 s against a nominal 21 s schedule.

Qualitative Comparison



Fig2. Clips confirmed by the pose-centric hybrid. clutching_chest and staggering were silently missed by both X3D-UGT variants.

Conclusion

DISCUSSION

- Recognition and interaction policy should be evaluated together but not conflated: any confirmed distress label begins the same confirmation, so what matters is whether a cue appears at all.
- Asking first turns a false trigger into a prompt rather than an alert, though it does not repair a silent miss.

LIMITATIONS AND FUTURE WORK

- 28 scripted clips from one actor; on-screen text prompts, not validated speech; no user study yet, and controller trials bypass the camera and pan-tilt execution.
- Next: a multi-participant end-to-end study, on-device profiling, and higher sensitivity on subtle upper-body cues.